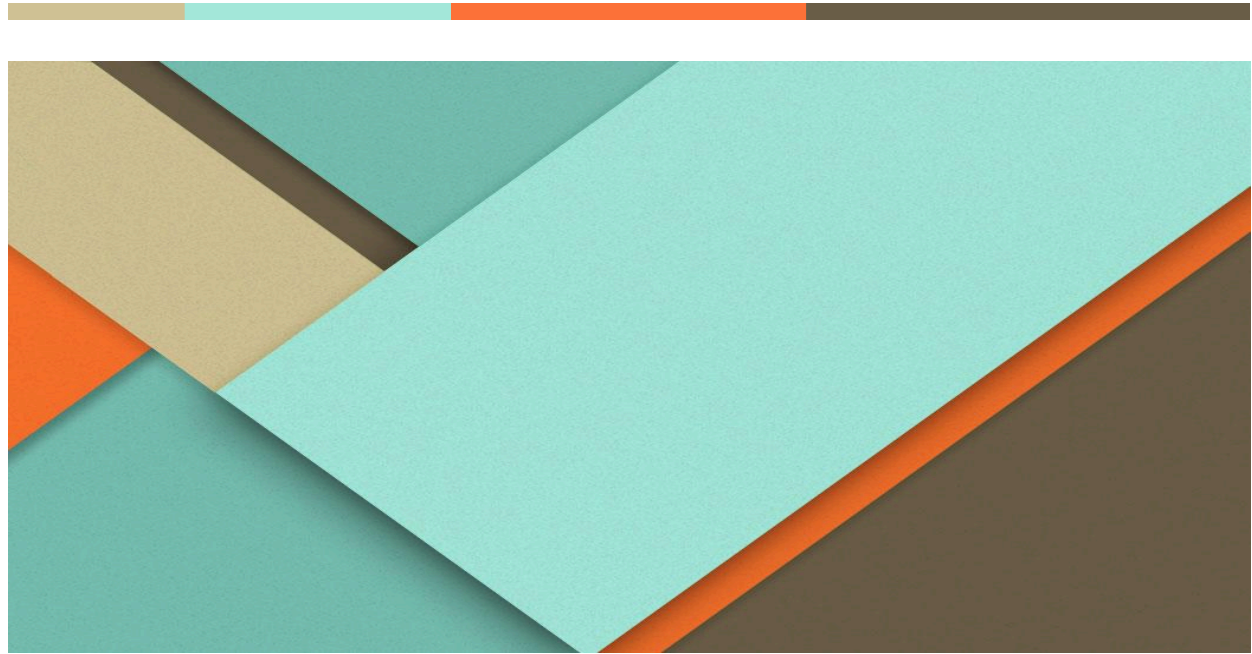




Consumer Online Purchase Behavior Study 2026

Submitted by:
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Executive Summary

This study analyzes consumer online purchasing behavior using primary survey data (n=113). Statistical tools in Microsoft Excel were used to examine platform preferences, the influence of income on purchase frequency, and brand loyalty patterns. Correlation and visualization techniques were applied to derive actionable marketing insights.

Introduction

With the rapid growth of e-commerce platforms such as Amazon, Flipkart, and Myntra, online purchasing has become a common consumer behavior. Income level is often considered a key factor influencing consumer buying decisions. This study aims to examine whether income significantly influences online purchase frequency among consumers.

Literature Review

Growth of Online Shopping in India:

Over the past decade, India has witnessed rapid growth in e-commerce due to increased internet penetration, smartphone adoption, digital payments, and improved logistics infrastructure. Platforms such as Amazon, Flipkart, and Myntra have transformed consumer buying behavior by offering convenience, variety, and competitive pricing. According to industry reports, online retail sales in India continue to grow at double-digit rates annually. This expansion has reshaped traditional purchasing patterns and increased consumer reliance on digital platforms.

Factors Influencing Online Purchase Behavior:

Consumer purchase behavior in online environments is influenced by multiple factors, including price sensitivity, convenience, trust, product variety, user experience, and promotional offers. Psychological factors such as perceived risk, brand reputation, and peer influence also play significant roles. Research in consumer behavior suggests that buying decisions are rarely driven by a single factor; instead, they result from a combination of economic, social, and behavioral variables.

Income as a Determinant of Purchasing Decisions:

Income is traditionally considered an important economic determinant of purchasing power. Higher income levels often enable consumers to make more frequent purchases and choose premium products. However, in digital markets, income may not always be the strongest predictor of purchase frequency. Factors such as discounts, seasonal sales, and easy payment options may reduce the impact of income constraints.

Brand Loyalty in the Digital Era:

Brand loyalty in online markets is influenced by customer satisfaction, delivery reliability, return policies, and reward programs. Younger consumers, particularly those between 18–35 years, tend to demonstrate stronger digital engagement and loyalty toward preferred platforms. Loyalty programs, personalized recommendations, and subscription models are increasingly used to retain customers.

Objectives of the Study

- To analyze the relationship between income and purchase frequency.
- To measure the correlation between income level and online buying behavior.
- To provide insights for e-commerce marketing strategies.

Research Methodology

- Data Type: Primary Data
- Data Collection Tool: Google Form
- Sample Size: 113 respondents
- Tool Used for Analysis: Microsoft Excel
- Statistical Techniques: Percentage Analysis, Correlation, Trendline (R^2)
- Research Design: Descriptive and Analytical

Data Description

The dataset used in this study consists of 113 responses collected through an online survey. The questionnaire included multiple questions related to consumer demographics, online shopping habits, platform preferences, brand loyalty, and factors influencing purchasing decisions.

Key variables analyzed in this research include:

- Monthly Income Level
- Online Purchase Frequency
- Preferred E-commerce Platform
- Brand Loyalty Behavior
- Age Group

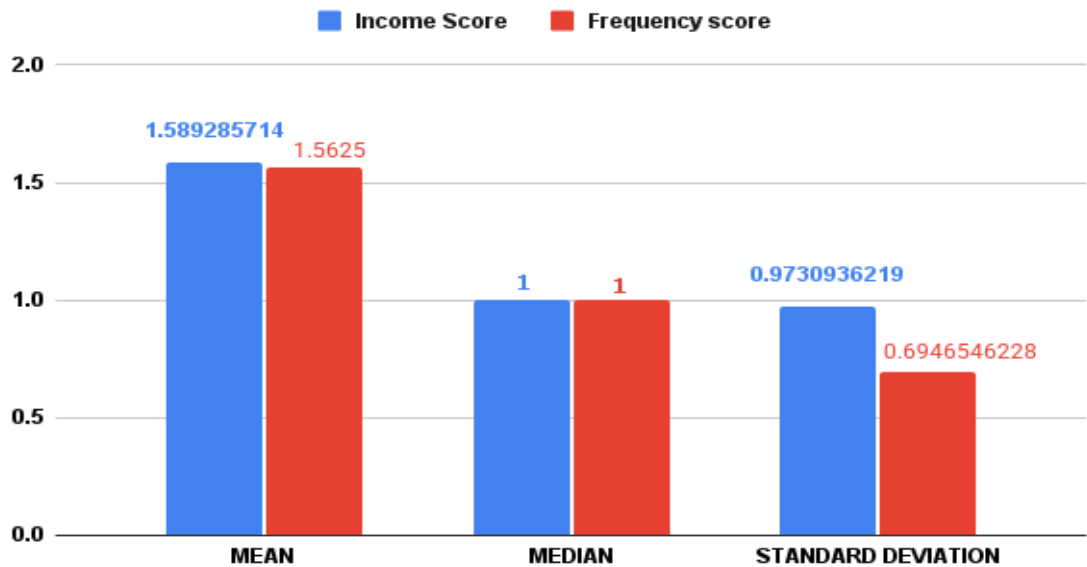
These variables were coded into numerical values to enable statistical analysis using Microsoft Excel.

Descriptive Statistics

Table 1: Descriptive Statistics of Income and Purchase Frequency Table

Statistic	Income Score	Frequency Score
Mean	1.589285714	1.5625
Median	1	1
Std. Deviation	0.9730936219	0.6946546228

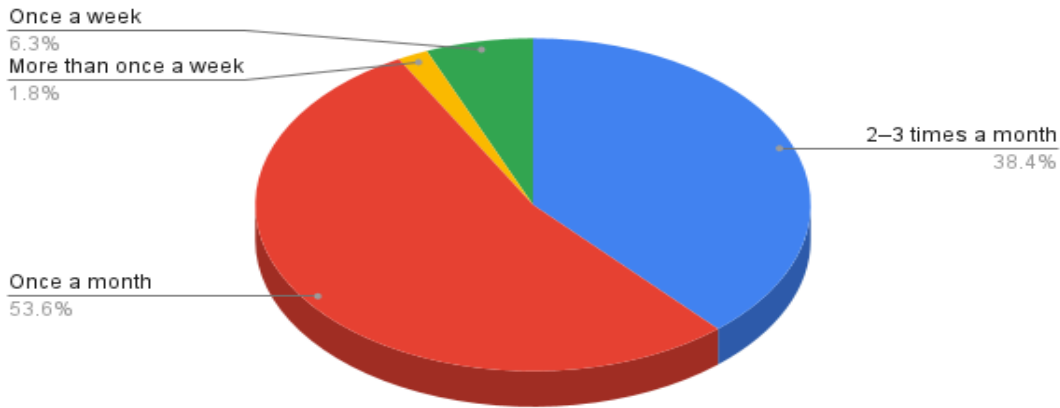
Figure 1: Descriptive Statistics of Income and Purchase Frequency



The descriptive statistics provide insights into the distribution of respondents' income levels and their online purchase frequency. The mean value represents the average score of respondents within the coded scale, while the median indicates the central tendency of the dataset. The standard deviation measures the variability of responses among participants. A moderate standard deviation suggests that responses are reasonably spread across different income groups and purchasing frequencies.

Figure 2: Online Purchase Frequency Among Respondents

Online Purchase Frequency Among Respondents

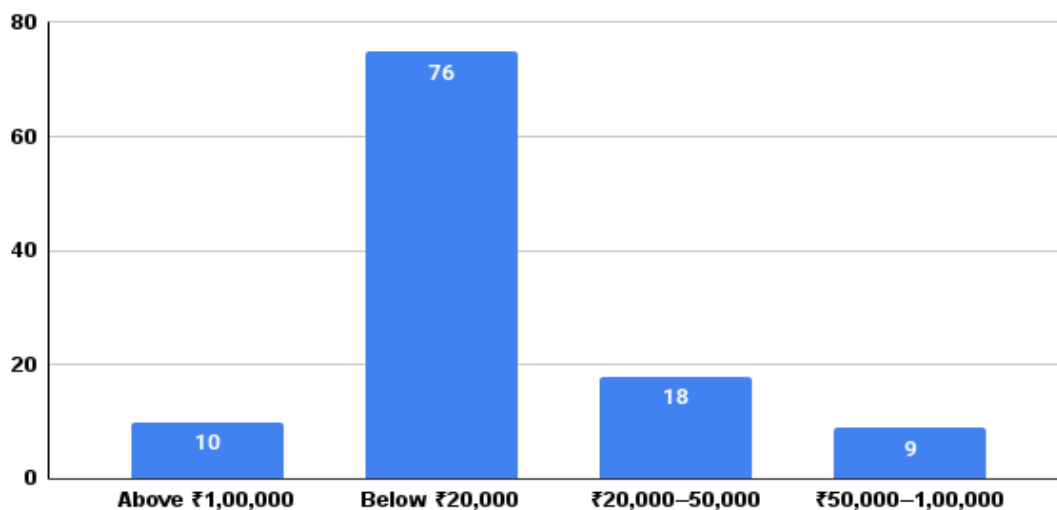


The above chart illustrates how frequently respondents purchase products online. The majority of respondents reported purchasing **once a month**, followed by **2-3 times a month**. Only a small proportion of respondents reported purchasing **once a week or more than once a week**.

This indicates that while online shopping is common among respondents, extremely frequent purchases are relatively limited.

Figure 3: Distribution of Respondents by Monthly Income Level

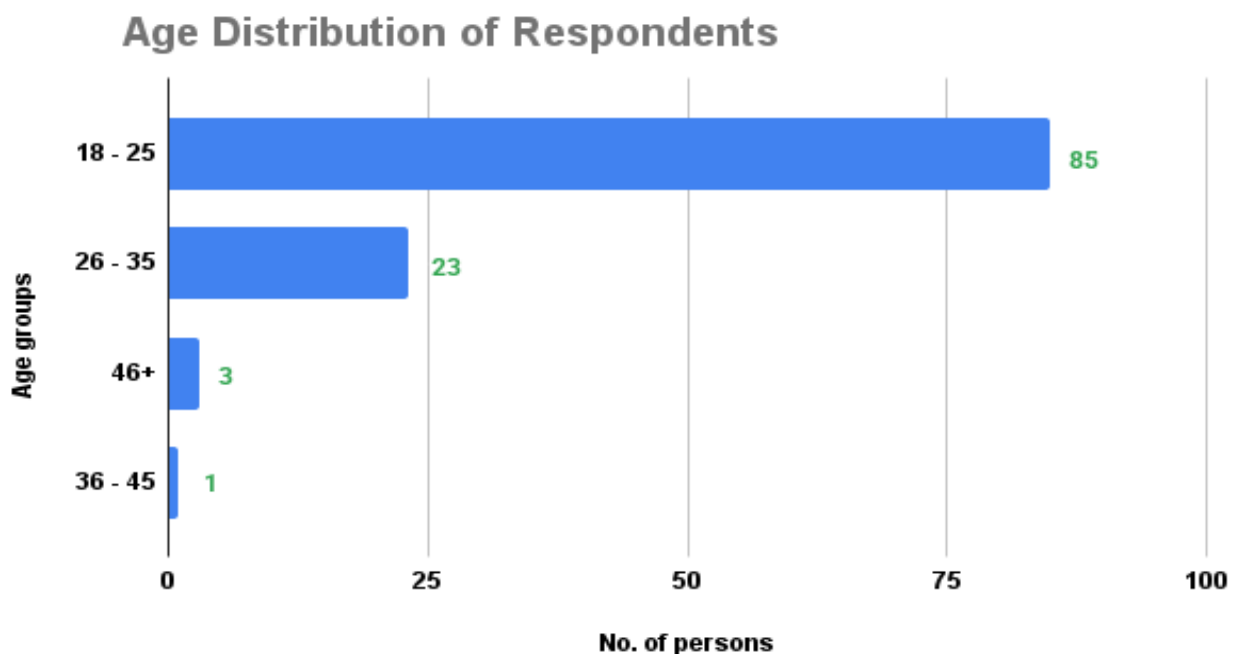
Distribution of Respondents by Monthly Income Level



The above chart shows the distribution of respondents based on their monthly income levels. The majority of respondents fall in the **below ₹20,000 income category**, indicating that the sample contains a large proportion of students or early-career individuals. A smaller proportion belongs to higher income brackets such as **₹20,000–₹50,000**, **₹50,000–₹1,00,000**, and **above ₹1,00,000**.

This income distribution is important for analyzing whether income significantly influences online purchasing behavior.

Figure 4: Age Distribution of Respondents



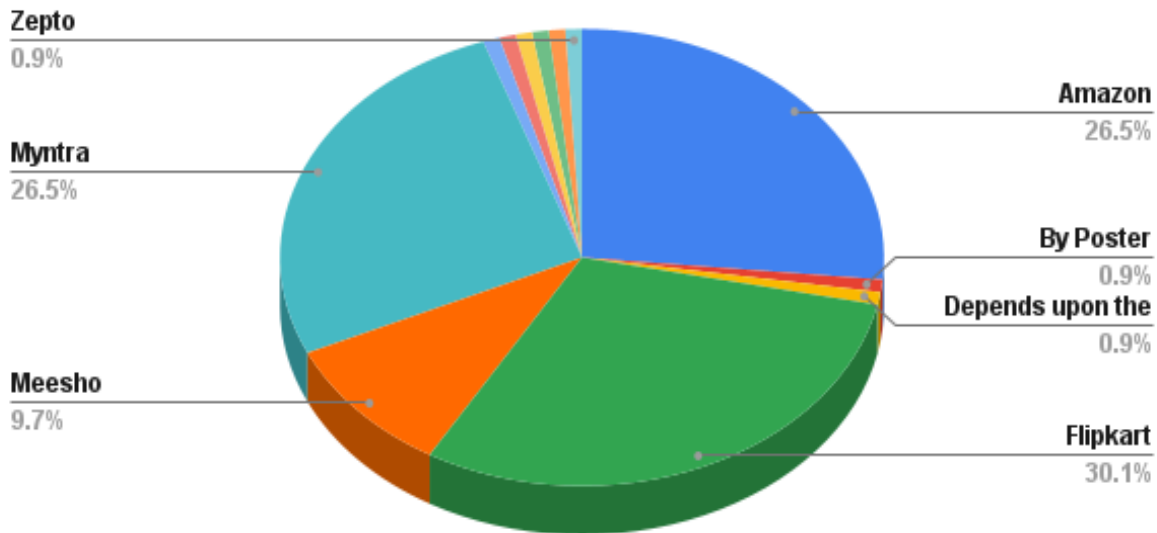
The above chart shows the distribution of respondents across different age groups. The majority of respondents belong to the 18–25 age group, followed by the 26–35 category. A smaller proportion of respondents belong to older age groups such as 36–45 and 46+.

This indicates that younger consumers are more actively participating in online shopping surveys. Since younger age groups are more digitally engaged, e-commerce platforms often focus their marketing strategies on these segments.

Platform Preference Analysis

Figure 4: Platform Preference Distribution (n=113)

Platform Preference Distribution (n=113)



The chart shows the distribution of respondents across different e-commerce platforms. Flipkart and Amazon have the highest share among respondents. Myntra also shows significant usage. This indicates intense competition among major e-commerce platforms.

Flipkart (30.1%) and Amazon (26.5%) dominate platform preference, followed closely by Myntra (26.5%). Meesho holds 9.7%, while smaller platforms contribute marginal shares. This indicates strong competition among major e-commerce platforms.

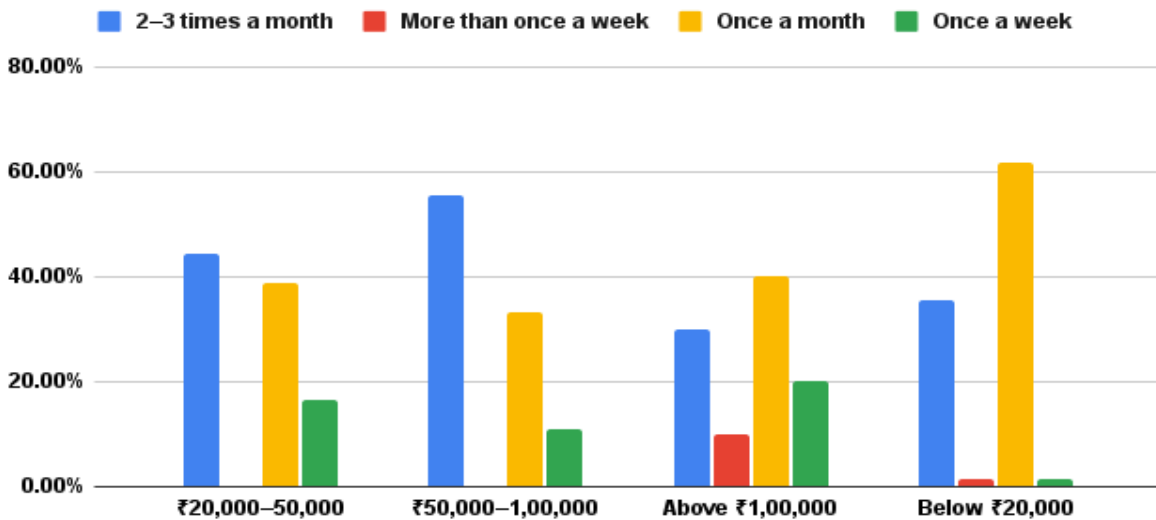
Business Insight:

Competitive differentiation strategies are critical in saturated platform markets.

Income vs Monthly Shopping Frequency

Figure 5: Monthly Income vs Online Shopping Frequency (% Distribution)

Monthly Income vs Online Shopping Frequency (% Distribution)



Middle-income groups (₹50,000–1,00,000) show the highest 2-3 times monthly purchase behavior. Lower income groups tend to purchase online less frequently, typically once per month.

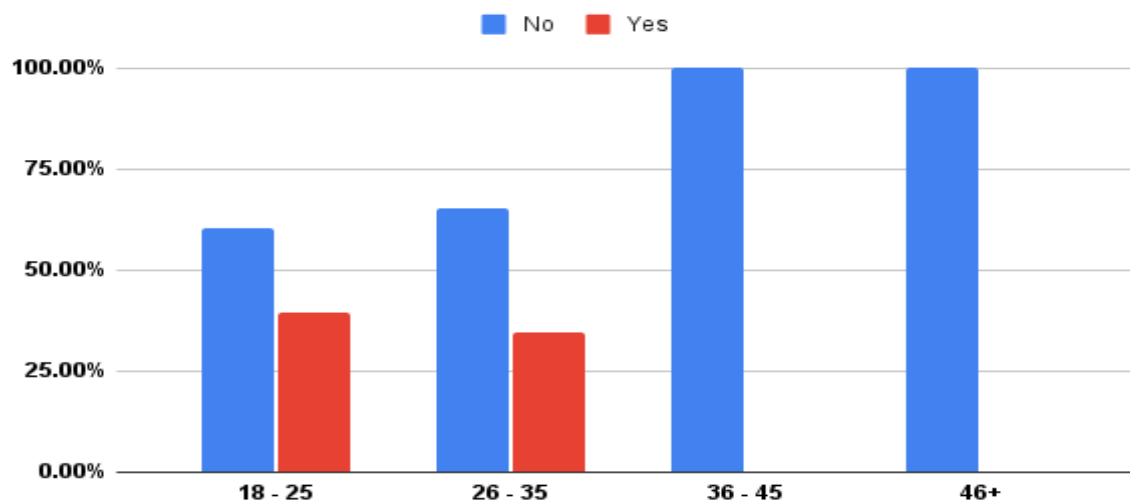
Business Insight:

Income influences purchase frequency but does not drastically change behavior patterns.

Age Group vs Brand Loyalty

Figure 6: Age Group vs Brand Loyalty Distribution

Age Group vs Brand Loyalty(% Distribution)



Younger consumers (18–35) exhibit relatively higher brand loyalty compared to older groups (36+), who demonstrate lower levels of loyalty.

Business Insight:

Loyalty programs should target young demographics.

Data Coding Section

Figure 7: Numerical Coding of Income and Purchase Frequency

9. Income Score	10. Frequency Score
4	2
1	2
1	1
1	1
1	4
2	1
1	2
1	2
1	1
4	2

To facilitate quantitative analysis, income categories and purchase frequency responses were converted into numerical scores using a 1–4 scale.

Income categories were converted into numerical scores:

- 1 = Below ₹20,000
- 2 = ₹20,000–50,000
- 3 = ₹50,000–1,00,000
- 4 = Above ₹1,00,000

Purchase frequency was also converted into a 1–4 scale:

- 1 = Once a month
- 2 = 2–3 times a month
- 3 = Once a week
- 4 = More than once a week

Correlation & Scatter Plot Analysis

The correlation between income score and purchase frequency score was calculated using the CORREL function in Microsoft Excel.

Formula:

=CORREL(J2:J114,K2:K114)

Result:

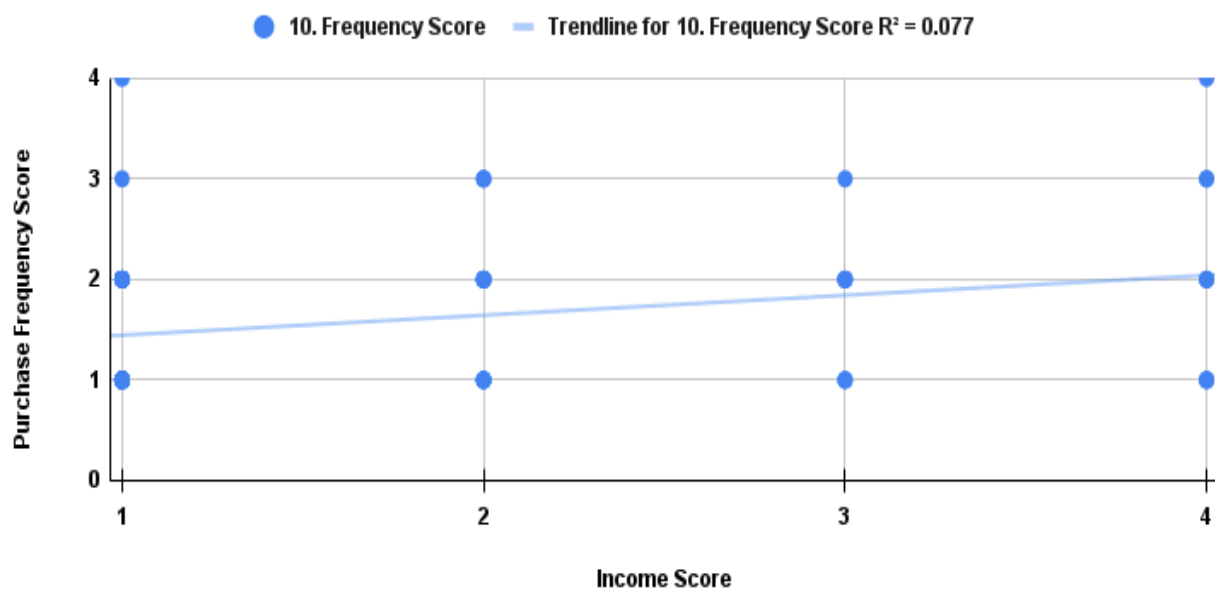
$r = 0.27$ (approx.)

Interpretation:

The correlation value of 0.27 indicates a weak positive relationship between income and online purchase frequency.

Figure 8: Scatter Plot Showing Income vs Online Purchase Frequency

Income vs Online Purchase Frequency



The correlation coefficient ($r \approx 0.27$) indicates a weak positive relationship. The R^2 value of 0.077 suggests that income explains only 7.7% of the variance in purchase frequency.

Business Interpretation:

Income has limited predictive power; marketing efforts should focus on behavioral and psychological factors.

Regression Analysis

A simple linear regression model was estimated to examine the effect of income on online purchase frequency.

Model specification:

$$\text{Purchase Frequency} = \beta_0 + \beta_1 (\text{Income Score}) + \varepsilon$$

Income demonstrates limited predictive power in explaining online purchasing behavior.

The regression coefficient ($\beta = 0.198$) suggests a weak positive relationship between income and purchase frequency. This means that for every one-unit increase in income score, purchase frequency increases by approximately 0.198 units.

The intercept value ($\beta_0 = 1.246$) represents the baseline purchase frequency when income score is zero.

Overall, the results indicate that income alone is not a strong predictor of online purchasing behavior.

Table 2: Regression Results Showing Effect of Income on Online Purchase Frequency

Regression Component	Value
Intercept (β_0)	1.246
Income Coefficient (β_1)	0.198
R^2 (Coefficient of Determination)	0.077
Interpretation	Weak positive relationship

A simple linear regression analysis was conducted to examine the effect of income on online purchase frequency. The regression coefficient for income score ($\beta = 0.198$) indicates a positive but weak relationship between income and purchase frequency. The R^2 value of 0.077 suggests that income explains only 7.7% of the variation in online purchasing behavior among respondents.

The intercept value ($\beta_0 = 1.246$) represents the baseline purchase frequency when the income score is zero. Overall, the regression model indicates that income alone is not a strong predictor of online purchase frequency.

The regression equation derived from the model can be expressed as:

$$\text{Purchase Frequency} = 1.246 + 0.198 (\text{Income Score})$$

This equation indicates that purchase frequency increases slightly with higher income levels, although the effect remains weak.

Managerial Implications

The findings of this study provide several important insights for e-commerce platforms and marketing managers. First, the analysis shows that income alone is not a strong predictor of online purchase frequency. This suggests that companies should not rely solely on income-based segmentation when targeting customers.

Second, behavioral factors such as convenience, discounts, and product variety may play a larger role in influencing online shopping behavior. Therefore, marketers should focus on personalized promotions and targeted advertising strategies.

Third, loyalty programs and personalized recommendations can help strengthen brand engagement among younger consumers who are highly active in online shopping.

Finally, companies should invest in improving customer experience, delivery reliability, and return policies, as these factors significantly influence repeat purchase behavior in digital markets.

Limitations of the Study

Although the study provides useful insights into consumer online purchase behavior, several limitations should be acknowledged. First, the sample size of 113 respondents may not fully represent the entire population of online consumers.

Second, the study relies on self-reported survey data, which may contain response bias or inaccuracies in reporting purchasing behavior.

Third, the research focuses primarily on income as a determinant of purchase frequency, while other important variables such as education, occupation, and lifestyle were not included in the analysis.

Fourth, the survey responses were collected over a short period of time, which limits the ability to analyze long-term consumer behavior trends.

Finally, the geographic distribution of respondents may not fully capture regional differences in online shopping behavior across India.

Scope for Future Research

Future research can expand this study by including additional variables such as age, education level, occupation, and digital literacy to better understand consumer purchase behavior.

Researchers may also conduct studies with larger sample sizes and broader geographic coverage to improve the generalizability of the findings.

Advanced statistical techniques such as multiple regression analysis and structural equation modeling could provide deeper insights into the relationships between different consumer behavior factors.

Longitudinal studies may also help track how online purchase behavior evolves over time as digital platforms continue to expand.

Finally, future studies could analyze platform-specific consumer behavior for major e-commerce companies to understand brand loyalty and competitive dynamics in the online retail market.

Conclusion

This study aimed to analyze the relationship between income level and online purchase frequency among consumers. Using survey data collected from 113 respondents, the research applied descriptive statistics, correlation analysis, and regression analysis to examine the relationship between these variables.

The results of the correlation analysis revealed a weak positive relationship between income and purchase frequency. Similarly, the regression analysis indicated that income explains only a small portion of the variation in online purchasing behavior.

These findings suggest that factors other than income, such as convenience, discounts, product availability, and customer experience, may play a more significant role in influencing online shopping behavior.

Overall, the study highlights the importance of adopting data-driven marketing strategies that focus on consumer behavior rather than relying solely on traditional demographic variables.

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